

Valfix Film Coated Tablet 150mg

Ingredient(s):
Each tablet contains:
Lamivudine 150mg

Pharmacology (Summary of Pharmacodynamic and Pharmacokinetic):

Pharmacodynamic:

Lamivudine is a nucleoside analogue which has activity against human immunodeficiency virus (HIV) and hepatitis B virus (HBV). It is metabolised intracellularly to the active moiety, lamivudine 5'-triphosphate. Its main mode of action is as a chain terminator of viral reverse transcription. The triphosphate has selective inhibitory activity against HIV-1 and HIV-2 replication *in vitro*, it is also active against zidovudine-resistant clinical isolates of HIV. No antagonistic effects *in vitro* were seen with lamivudine and other anti retrovirals (tested agents: abacavir, didanosine, nevirapine and zidovudine).

Pharmacokinetic:

Absorption

Lamivudine is well absorbed from the gastrointestinal tract, and the bioavailability of oral lamivudine in adults is normally between 80 and 85%. Following oral administration, the mean time (t_{max}) to maximal serum concentrations (C_{max}) is about an hour.

Co-administration of lamivudine with food results in a delay of t_{max} and a lower C_{max} . However, the extent (based on the AUC) of lamivudine absorbed is not influenced.

Administration of crushed tablets with a small amount of semi-solid food or liquid would not be expected to have an impact on the pharmaceutical quality, and would therefore not be expected to alter the clinical effect. This conclusion is based on the physicochemical and pharmacokinetic data assuming that the patient crushes and transfers 100% of the tablet and ingests immediately.

Distribution

The mean volume of distribution is 1.3 l/kg. The observed half-life of elimination is 5 to 7 hours. The mean systemic clearance of lamivudine is approximately 0.32 l/h/kg, with predominantly renal clearance (> 70%) via the organic cationic transport system.

Lamivudine exhibits linear pharmacokinetics over the therapeutic dose range and displays limited binding to the major plasma protein albumin.

Biotransformation

The active moiety, intracellular lamivudine triphosphate, has a prolonged terminal half-life in the cell (16 to 19 hours) compared to the plasma lamivudine half-life (5 to 7 hours).

Lamivudine is predominately cleared unchanged by renal excretion. The likelihood of metabolic interactions of lamivudine with other medicinal products is low due to the small extent of hepatic metabolism (5-10%) and low plasma protein binding.

Elimination

Lamivudine elimination is affected by renal dysfunction. A recommended dosage regimen for patients with creatinine clearance below 50 ml/min is shown in the dosage section.

Indication(s):

In combination with other antiretroviral agents for treating HIV infected adults & children.

Dosage and Administration

Adults and adolescents (weighing at least 30 kg):

150mg twice daily or 300 mg once daily, to be taken with or without food.

Children aged more than 3 months:

Children weighing between 21kg to 30kg: 75mg taken in the morning and 150mg taken in the evening.

Children weighing between 14kg to 21kg: 75mg twice daily.

Children less than three months of age: The limited data available are insufficient to propose specific dosage recommendations.

Special populations:

Elderly: No specific data are available; however, special care is advised in this age group due to age-associated changes such as the decrease in renal function and alteration of haematological parameters.

Renal impairment: Lamivudine plasma concentrations (AUC) are increased in patients with moderate - severe renal impairment due to decreased clearance. The dosage should therefore be reduced for patients with a creatinine clearance of less than 50ml/min as shown in Table 1 and Table 2. The same percentage reduction in dose applies for paediatric patients with renal impairment.

Table 1: Dosing recommendations – Adults and adolescents weighing at least 30kg:

When doses below 150mg are required, other approved dosage forms and strengths of Lamivudine should be used in such cases.

Creatinine clearance (ml/min)	First dose	Maintenance dose
30 to <50	150 mg	150 mg once daily
15 to <30	150 mg	100 mg once daily
5 to <15	150 mg	50 mg once daily
<5	50 mg	25 mg once daily

Table 2: Dosing recommendations – Children aged more than 3 months and weighing less than 30 kg:

Creatinine clearance (ml/min)	First dose	Maintenance dose
30 to <50	4 mg/kg	4 mg/kg once daily
15 to <30	4 mg/kg	2.6 mg/kg once daily
5 to <15	4 mg/kg	1.3 mg/kg once daily
<5	1.3 mg/kg	0.7 mg/kg once daily

Hepatic impairment: No dose adjustment is necessary in patients with moderate or severe hepatic impairment unless accompanied by renal impairment.

Mode of Administration:

To be taken orally with or without food. To ensure administration of the entire dose, the tablet(s) should ideally be swallowed without crushing.

Alternatively, for patients who are unable to swallow tablets, the tablet(s) may be crushed and added to a small amount of semi-solid food or liquid, all of which should be consumed immediately.

Contraindication:

The use of Lamivudine Tablet is contraindicated in patients with known hypersensitivity to lamivudine or to any ingredient of the preparation.

Warning and Precautions:

- While effective viral suppression with antiretroviral therapy has been proven to substantially reduce the risk of sexual transmission, a residual risk cannot be excluded. Precautions to prevent transmission should be taken.
- Lamivudine is not recommended for use as monotherapy.
- Renal impairment:** In patients with moderate to severe renal impairment, the terminal plasma half-life of lamivudine is increased due to decreased clearance, therefore the dose should be adjusted.
- Triple nucleoside therapy:** There have been reports of a high rate of virological failure and of emergence of resistance at an early stage when lamivudine was combined with tenofovir disoproxil fumarate and abacavir as well as with tenofovir disoproxil fumarate and didanosine as a once daily regimen.
- Opportunistic infections:** Patients receiving lamivudine or any other antiretroviral therapy may continue to develop opportunistic infections and other complications of HIV infection, and therefore should remain under close clinical observation by physicians experienced in the treatment of patients with associated HIV diseases.
- Pancreatitis:** Cases of pancreatitis have occurred rarely. However, it is not clear whether these cases were due to the antiretroviral treatment or to the underlying HIV disease. Treatment with lamivudine should be stopped immediately if clinical signs, symptoms or laboratory abnormalities suggestive of pancreatitis occur.
- Mitochondrial dysfunction following exposure in utero:** Nucleoside and nucleotide analogues may impact mitochondrial function to a variable degree, which is most pronounced with stavudine, didanosine and zidovudine. There have been reports of mitochondrial dysfunction in HIV-negative infants exposed *in utero* and/or post-natally to nucleoside analogues; these have predominantly concerned treatment with regimens containing zidovudine. The main adverse reactions reported are haematological disorders (anaemia, neutropenia) and metabolic disorders (hyperlactatemia, hyperlipasemia). These events have often been transitory. Late-onset neurological disorders have been reported rarely (hypertonia, convulsion, abnormal behaviour). Whether such neurological disorders are transient or permanent is currently unknown. These findings should be considered for any child exposed *in utero* to nucleoside and nucleotide analogues, who presents with severe clinical findings of unknown aetiology, particularly neurological findings.

8. **Weight and metabolic parameters:** An increase in weight and in levels of blood lipids and glucose may occur during antiretroviral therapy. Such changes may in part be linked to disease control and life style. For lipids, there is in some cases evidence for a treatment effect, while for weight gain there is no strong evidence relating this to any particular treatment. For monitoring of blood lipids and glucose reference is made to established HIV treatment guidelines. Lipid disorders should be managed as clinically appropriate.
9. **Immune Reactivation Syndrome:** In HIV-infected patients with severe immune deficiency at the time of institution of combination antiretroviral therapy (CART), an inflammatory reaction to asymptomatic or residual opportunistic pathogens may arise and cause serious clinical conditions, or aggravation of symptoms. Typically, such reactions have been observed within the first few weeks or months of initiation of CART. Relevant examples are cytomegalovirus retinitis, generalised and/or focal mycobacterium infections, and *Pneumocystis carinii* pneumonia. Any inflammatory symptoms should be evaluated and treatment instituted when necessary. Autoimmune disorders (such as Graves' disease) have also been reported to occur in the setting of immune reactivation; however, the reported time to onset is more variable and these events can occur many months after initiation of treatment.
10. Patients with chronic hepatitis B or C and treated with combination antiretroviral therapy are at an increased risk of severe and potentially fatal hepatic adverse events. In case of concomitant antiviral therapy for hepatitis B or C, please refer also to the relevant product information for these medicinal products.
11. If lamivudine is discontinued in patients co-infected with hepatitis B virus, periodic monitoring of liver function tests and markers of HBV replication is recommended, as withdrawal of lamivudine may result in an acute exacerbation of hepatitis.
12. Patients with pre-existing liver dysfunction, including chronic active hepatitis, have an increased frequency of liver function abnormalities during combination antiretroviral therapy, and should be monitored according to standard practice. If there is evidence of worsening liver disease in such patients, interruption or discontinuation of treatment must be considered.
13. **Osteonecrosis:** Although the aetiology is considered to be multifactorial (including corticosteroid use, alcohol consumption, severe immunosuppression, higher body mass index), cases of osteonecrosis have been reported particularly in patients with advanced HIV-disease and/or long-term exposure to combination antiretroviral therapy (CART). Patients should be advised to seek medical advice if they experience joint aches and pain, joint stiffness or difficulty in movement.
14. **Effects on the ability to drive and use machines:** No studies on the effects on the ability to drive and use machines have been performed.

Interaction with Other Medicaments:

1. The likelihood of metabolic interactions is low due to limited metabolism and plasma protein-binding and almost complete renal clearance.
2. Administration of trimethoprim/sulfamethoxazole 160 mg/800 mg results in an increase in lamivudine exposure, because of the trimethoprim component; the sulfamethoxazole component did not interact. However, unless the patient has renal impairment, no dosage adjustment of lamivudine is necessary. Lamivudine has no effect on the pharmacokinetics of trimethoprim or sulfamethoxazole. When concomitant administration is warranted, patients should be monitored clinically. Co-administration of lamivudine with high doses of co-trimoxazole for the treatment of *Pneumocystis carinii* pneumonia (PCP) and toxoplasmosis should be avoided.
3. The possibility of interactions with other medicinal products administered concurrently should be considered, particularly when the main route of elimination is active renal secretion via the organic cationic transport system e.g. trimethoprim. Other medicinal products (e.g. ranitidine, cimetidine) are eliminated only in part by this mechanism and were shown not to interact with lamivudine. The nucleoside analogues (e.g. didanosine and zidovudine) are not eliminated by this mechanism and are unlikely to interact with lamivudine.
4. A modest increase in C_{max} was observed for zidovudine when administered with lamivudine, however overall exposure (AUC) is not significantly altered. Zidovudine has no effect on the pharmacokinetics of lamivudine.
5. Due to similarities, lamivudine should not be administered concomitantly with other cytidine analogues, such as emtricitabine. Moreover, lamivudine should not be taken with any other medicinal products containing lamivudine.
6. *In vitro* lamivudine inhibits the intracellular phosphorylation of cladribine leading to a potential risk of cladribine loss of efficacy in case of combination in the clinical setting. Some clinical findings also support a possible interaction between lamivudine and cladribine. Therefore, the concomitant use of lamivudine with cladribine is not recommended.
7. Lamivudine metabolism does not involve CYP3A, making interactions with medicinal products metabolised by this system (e.g. PIs) unlikely.

Pregnancy and Lactation:

Pregnancy

As a general rule, when deciding to use antiretroviral agents for the treatment of HIV infection in pregnant women and consequently for reducing the risk of HIV vertical transmission to the newborn, the animal data as well as the clinical experience in pregnant women should be taken into account.

Lamivudine can be used during pregnancy if clinically needed. The malformative risk is unlikely in humans.

For patients co-infected with hepatitis who are being treated with lamivudine and subsequently become pregnant, consideration should be given to the possibility of a recurrence of hepatitis on discontinuation of lamivudine.

Mitochondrial dysfunction: Nucleoside and nucleotide analogues have been demonstrated *in vitro* and *in vivo* to cause a variable degree of mitochondrial damage.

Breast-feeding

Following oral administration lamivudine was excreted in breast milk at similar concentrations to those found in serum. Serum concentrations of lamivudine in breastfed infants of mothers treated for HIV are very low and progressively decrease to undetectable levels when breastfed infants reach 24 weeks of age. There are no data available on the safety of lamivudine when administered to babies less than three months old. It is recommended that HIV infected women do not breast-feed their infants under any circumstances in order to avoid transmission of HIV.

Fertility

Lamivudine had no effect on fertility.

Side Effects:

The most common adverse events reported are hyperlactataemia, headache, nausea, vomiting, upper abdominal pain, diarrhoea, rash, alopecia, arthralgia, muscle disorders, fatigue, malaise and fever.

Weight and levels of blood lipids and glucose may increase during antiretroviral therapy.

In HIV-infected patients with severe immune deficiency at the time of initiation of combination antiretroviral therapy (CART), an inflammatory reaction to asymptomatic or residual opportunistic infections may arise. Autoimmune disorders (such as Graves' disease) have also been reported to occur in the setting of immune reactivation; however, the reported time to onset is more variable and these events can occur many months after initiation of treatment.

Cases of osteonecrosis have been reported, particularly in patients with generally acknowledged risk factors, advanced HIV disease or long-term combined antiretroviral exposure (CART). The frequency of which is unknown.

Symptoms and Treatment of Overdose:

Limited data are available on the consequences of ingestion of acute overdoses in humans. No fatalities occurred, and the patients recovered.

No specific signs or symptoms have been identified following such overdose.

If overdosage occurs the patient should be monitored, and standard supportive treatment applied as required. Since lamivudine is dialysable, continuous haemodialysis could be used in the treatment of overdosage, although this has not been studied.

Storage Condition:

Store at temperature below 30°C.

Shelf Life:

3 years from the date of manufacture.

Product Description:

A white colour film coated diamond shaped tablet, one side impressed with "YSP" 

Dosage Forms and Packaging available:

Blister packing of 10's x 3, 10's x 6 and 10's x 12.



Manufacturer and Product Registration Holder:
Y.S.P. INDUSTRIES (M) SDN. BHD. (199001001034)
 Lot 3, 5, 7 & 12, Jalan P/7, Section 13,
 Kawasan Perindustrian Bandar Baru Bangi,
 43000 Kajang, Selangor, Malaysia.
 Ordering Line: 1 800 88 3027
 Product Info: 1 800 88 3679